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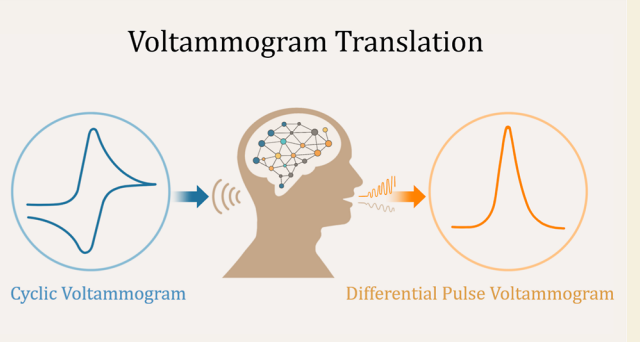
Voltammetric Translation with Transformers: Converting Cyclic Voltammograms to Differential Pulse Voltammograms

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ABSTRACT: Here we report a voltammogram translation framework that converts cyclic voltammograms into differential pulse voltammograms, enabling protocol interoperability without requiring paired experimental corpora. We formulate CV → DPV conversion as a sequence-to-sequence regression problem and implement a query-based transformer that encodes the full CV cycle with explicit scan-direction context and decodes DPV on a fixed target potential grid via cross-attention. To overcome the scarcity of paired measurements, we generate a large synthetic paired CV−DPV data set using physics-informed simulations spanning six representative mechanisms (E, EC, ECatal, EE, ECE, and ECEC) and broad parameter variations. Across deep-learning and classical regression baselines, the Transformer achieves the lowest prediction error and markedly outperforms slope-based heuristics. Cross-attention analysis indicates a two-mode strategy: baseline anchoring away from faradaic features and protocol-consistent local retrieval in peak-forming regions, including an offset comparable to the DPV pulse amplitude. Importantly, the experimental validation directly probes mechanism coverage learned from simulation: the translator accurately reproduces DPV for *p*-benzoquinone as an ECEC system and produces a virtual DPV that tracks peak evolution during aniline electropolymerization, a regime where direct DPV acquisition is often impractical.



KEYWORDS: *voltammogram conversion, cyclic voltammetry, differential pulse voltammetry, transformer neural network, cross-attention interpretability, electrochemical reaction mechanism classification*

1. INTRODUCTION

Voltammetry underpins modern electrochemical analysis[1,2] because a prescribed potential waveform encodes kinetics, transport, and interfacial processes into current responses that are directly useful for sensing,[3] catalyst screening,[4] energy materials,[5] and mechanistic studies.[6] The field therefore relies on a family of waveform protocols—including cyclic voltammetry (CV),[7,8] square-wave voltammetry (SWV),[9,10] normal pulse voltammetry (NPV)[11] and differential pulse voltammetry (DPV)[12]—that interrogate the same electrochemical system through different excitation patterns and consequently trade off sensitivity, baseline suppression, and interpretability.[2,13] Importantly, protocol choice is not a mere procedural detail: pulse-derived methods (NPV, SWV, DPV) can sharpen faradaic features and reduce capacitive backgrounds,[2,9−12] whereas CV preserves scan-history information across a full cycle.[7,8] Yet this protocol diversity also creates a practical interoperability challenge, as data sets acquired under different protocols are difficult to compare directly and analysis pipelines often become protocol-specific.[6,14]

This raises a natural question: can one learn a mapping between voltammograms measured under different protocols? A useful analogy is language translation. Different voltammetric protocols can be viewed as distinct "dialects" that express the same underlying electrochemical information in different waveform forms; translation, in this context, means converting one waveform representation into another under comparable electrochemical conditions. Recent advances in neural machine translation have established Transformer-based sequence-to-sequence models as effective translators because they learn context-dependent mappings while capturing long-range dependencies through attention.[15−17] The same modeling advantage is attractive for voltammetric signals, where local features (e.g., onset, peak shape, baseline curvature) can depend on broader context across the potential window and, for cyclic measurements, on scan history.

Here we show that voltammogram translation—schematized in Scheme 1—enables protocol-to-protocol conversion by casting the task as conditional sequence-to-sequence regression,

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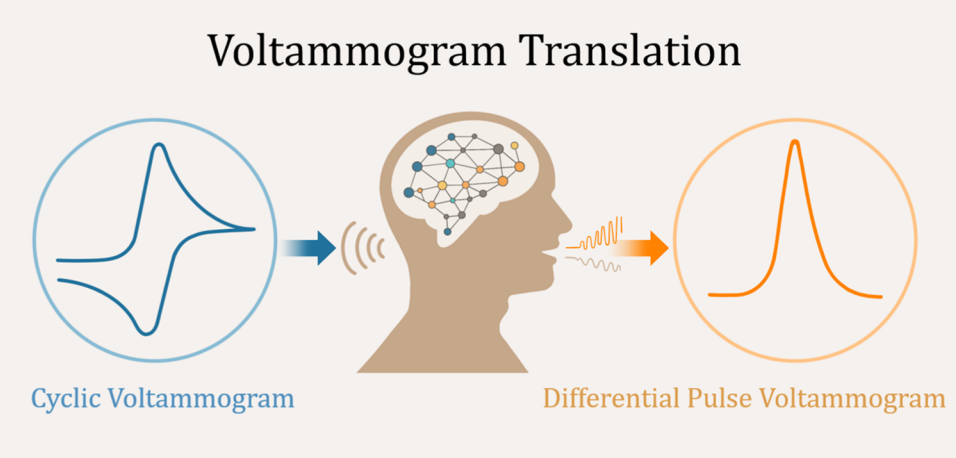


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Scheme 1. Concept of Voltammogram Translation: a Cyclic Voltammogram (CV) Is Provided as the Input and Mapped by the Proposed Neural-Network Model to the Corresponding Differential Pulse Voltammogram (DPV) as the Output



using CV → DPV as a proof-of-concept within the voltammetry family. We focus on DPV because its response typically exhibits more structured and nontrivial waveform characteristics than SWV or NPV under comparable conditions, providing a more demanding target for protocol translation.[2,7,9,10,12,18] We introduce a query-based transformer that represents the full CV cycle as continuous potential−current tokens with scan-context encoding and generates DPV values on a fixed target potential grid. Although we use CV → DPV as a representative case, the same translation formulation is intended to extend to other protocol pairs (e.g., involving SWV or NPV) given appropriate paired data and protocol specifications.

Because large, well-paired experimental corpora spanning multiple voltammetric protocols are difficult to obtain, we generate diverse paired data sets using physics-informed simulations under controlled variations in electrochemical parameters and measurement settings.[19−21] This design allows us to evaluate feasibility in a controlled setting and to test whether a Transformer can learn protocol-dependent waveform transformations without protocol-specific hand-crafted rules. More broadly, a practical translator between voltammograms could support cross-protocol data set harmonization, systematic augmentation of scarce paired measurements, and informed selection of follow-up protocols, while remaining complementary to mechanistic interpretation and first-principles simulation.

In this report, we first define the translation task and model formulation, then describe data generation and training, and finally report benchmarking, ablations, robustness tests, and experimental validation.

2. THEORY

2.1. Problem Definition: CV → DPV Waveform Translation

We formulate voltammogram translation, conceptualized in Scheme 1, as conditional waveform regression from a cyclic voltammogram (CV) to a differential pulse voltammogram (DPV), under matched electrochemical conditions and protocol specifications. A CV is treated as an ordered sequence of sampled potential−current observations across the full cycle (forward and return scans). Let

$$\mathbf{x} = \{(E_i, I_i)\}_{i=1}^{S}$$

where E_i denotes the applied potential and I_i the corresponding measured current at the i-th sampling point and S is the number of sampled points in the CV cycle. The DPV target is defined on a prescribed potential grid $\{E_j^*\}_{j=1}^{T}$, (typically corresponding to a single sweep direction in the DPV protocol). We denote the DPV waveform as

$$\mathbf{y} = \{y_j\}_{j=1}^{T}, \quad y_j = I_{\mathrm{DPV}}(E_j^*)$$

where T is the number of DPV grid points. Given paired measurements $\{(\mathbf{x}^{(n)}, \mathbf{y}^{(n)})\}_{n=1}^{N}$, the objective is to learn a parametric mapping f_θ such that

$$\hat{\mathbf{y}} = f_\theta(\mathbf{x}), \quad \hat{\mathbf{y}} \in \mathbb{R}^T$$

and to minimize a regression loss between $\hat{\mathbf{y}}$ and $\mathbf{y}$ over the data set.

Importantly, our goal is not mechanistic inference (e.g., estimating kinetic parameters or identifying reaction pathways). Instead, we treat CV → DPV as a protocol-defined operator that translates one measurement modality into another. This perspective motivates (i) representing voltammograms as structured sequences and (ii) using attention-based conditional modeling to capture context-dependent relationships across the potential window and across scan history.[15−17]

2.2. CV Tokenization with Cycle Awareness: Embedding, Position, and Scan-Branch Context

Transformers operate on token sequences. We therefore represent each CV sampling point as a continuous-valued token with a feature vector

$$\mathbf{u}_i = [E_i, I_i, (\mathrm{d}I/\mathrm{d}E)_i] \in \mathbb{R}^p$$

where $\mathrm{d}I/\mathrm{d}E$ is an optional auxiliary feature computed from the CV (when used), and p is the input dimension. Cyclic voltammetry contains forward and return scans, so similar potentials may appear twice with different history-dependent currents. To prevent the model from conflating these points, we include a scan-branch indicator $b_i \in 0,1$ and embed each token as

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{b_i}\mathbf{u}_i + \mathbf{e}_{b_i} + \mathbf{p}_i$$

where $\mathbf{W}_{bi}$ and $\mathbf{u}_i$ are direction-specific projection and embedding, and $\mathbf{p}_i$ is the positional encoding that preserves

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the order of the sampled sequence. These are minimal, model-native mechanisms that make a full-cycle CV a well-posed input sequence for attention-based modeling.

The Transformer encoder then produces contextualized representations ("memory") by multihead self-attention and feed-forward blocks

$$\mathbf{M} = \mathbf{Encoder}(\mathbf{H}^{(0)}), \quad \mathbf{M} \in \mathbb{R}^{S\times d}$$

where d is the model dimension and $\mathbf{M}$ aggregates information distributed across the CV cycle rather than relying on fixed local neighborhoods.

2.3. Query-Based DPV Prediction: Grid-Conditioned Cross-Attention

To predict a DPV waveform on a prescribed potential grid, we use a query-based decoder (Scheme 2). For each DPV grid point E_j^*, we form a query embedding

$$\mathbf{q}_j = \mathbf{W}_E \cdot E_j^* + \mathbf{q}_j^{\mathrm{lr}} + \mathbf{p}_j^*$$

where $\mathbf{W}_E$ is a linear projection from the scalar potential to $\mathbb{R}^d$, $\mathbf{q}_j^{\mathrm{lr}}$ is a learned query bias (one per grid position), and $\mathbf{p}_j^*$ is a positional encoding for the DPV grid. The model then computes cross-attention from the DPV queries to the encoded CV memory

$$\mathbf{z}_j = \mathrm{Attn}(\mathbf{q}_j, \mathbf{M}, \mathbf{M})$$

i.e., queries are $\mathbf{q}_j$ and keys/values are the CV memory $\mathbf{M}$. The predicted DPV value is obtained by a lightweight regression head:

$$\hat{y}_j = g(\mathbf{z}_j)$$

with $g(\cdot)$ implemented as a linear regressor on normalized latent features. The full DPV waveform $\hat{\mathbf{y}}$ is produced in parallel from the set of grid-conditioned queries, while each DPV point remains explicitly tied to its coordinate E_j^*.

Conceptually, this is the operational meaning of "translation" in our setting: the model learns an alignment-and-aggregation rule that selects which regions of the CV (possibly including scan-history cues from the return direction) are most informative for reconstructing each DPV point on the target potential grid.

2.4. Learning Objective and Scope of the Theory-Level Commitments

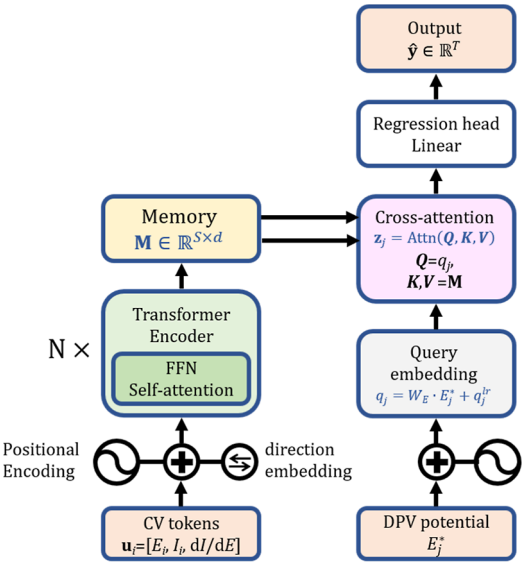
Because the output is a real-valued current waveform (not discrete tokens), training minimizes a continuous regression loss over paired waveforms

$$\min_{\theta} \frac{1}{N}\sum_{n=1}^{N} \mathcal{L}(f_\theta(\mathbf{x}^{(n)}), \mathbf{y}^{(n)})$$

where $\mathcal{L}$ is a pointwise waveform loss (MSE as the base objective). The theory-level commitments of this work are therefore

- Protocol-defined mapping: CV → DPV is treated as a translation operator between measurement modalities, not as mechanistic identification.
- Cycle-aware representation: full-cycle CV is encoded with explicit scan-branch context so that forward and return information is not conflated.
- Grid-conditioned decoding: DPV is generated by cross-attention from DPV potential queries to CV memory, producing the entire waveform on a prescribed grid (Scheme 2).

Scheme 2. Architecture of the Proposed Voltammogram Transformer for CV-to-DPV Translation; the Input Cyclic Voltammogram is Tokenized as $\mathbf{u}_i = [E_i, I_i, (\mathrm{d}I/\mathrm{d}E)_i]$ and Encoded by a Transformer Encoder to form a Memory Matrix $\mathbf{M} \in \mathbb{R}^{S\times d}$. For Each DPV Potential E_j^*, a Query Embedding q_j is Generated and Applied as Q in Cross-Attention with $K, V = M$ to Obtain z_j, which is Mapped by a Linear Regression Head to the Output $\hat{\mathbf{y}} \in \mathbb{R}^T$



Implementation details (data set construction, normalization, and specific training settings) are described in Methods and the Supporting Inforamtion, while the empirical value of the representation choices (e.g., using full-cycle CV and including $\mathrm{d}I/\mathrm{d}E$ as an auxiliary input feature) is validated in Results and Discussion.

3. METHODS

This section outlines the workflow for data generation, model training, and evaluation of the proposed CV → DPV translator. Detailed settings and implementation procedures are provided in the Supporting Information (SI−S1 to S6).

3.1. Task Definition

Each data set sample is a paired instance $(\mathbf{x}, \mathbf{y})$, where $\mathbf{x}$ is a discretized cyclic voltammogram (CV) trace and $\mathbf{y}$ is a discretized differential pulse voltammogram (DPV) waveform on a fixed potential grid. To distinguish the forward and reverse scans of potential, we include a scan-direction indicator for each CV point. The DPV target $\mathbf{y}$ is treated strictly as a vector-valued regression target on the predetermined grid of potential, and the model predicts the entire DPV waveform on that grid.

In this task, the model is not used to infer reaction mechanisms or extract kinetic parameters. Rather, we evaluate it as a protocol-to-protocol waveform translator, assessing prediction accuracy under fixed preprocessing and a DPV potential grid.

3.2. Synthetic Paired Data Set, Voltammetric Protocols, and Data Representation

To train and evaluate the CV → DPV translator under controlled conditions, we construct a synthetic data set of paired CV−DPV samples generated by physics-based electrochemical simulations.[19,22,23] We consider a library of representative reaction schemes covering common voltammetric behaviors, including E, EC, ECatalcatalytic), EE, ECE, and ECEC. For each sample, kinetic and transport parameters

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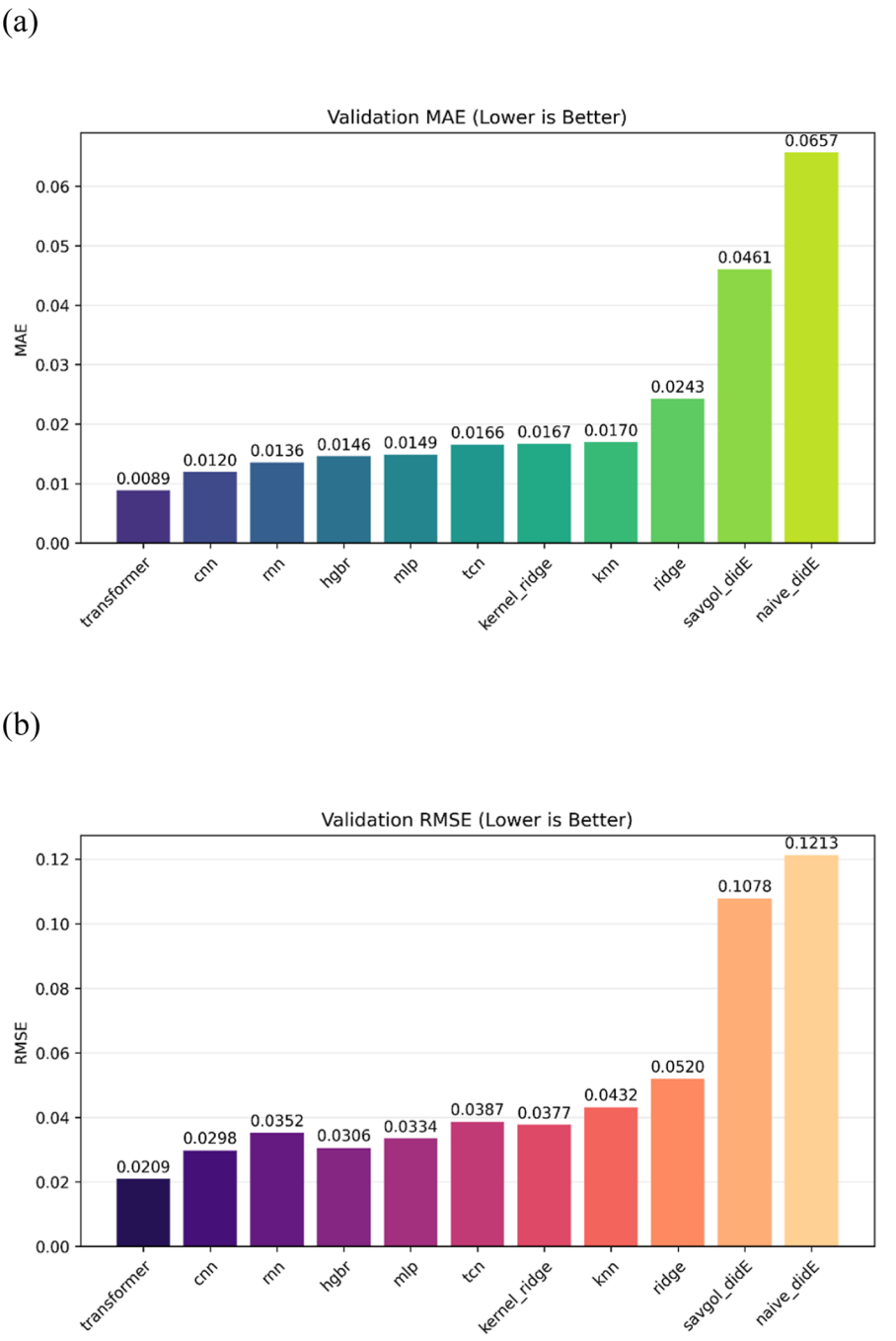


Figure 1. Validation performance comparison for CV−to−DPV translation across the proposed transformer and baseline methods. (a) Mean absolute error (MAE) and (b) root-mean-square error (RMSE) are evaluated on the validation set (lower is better), showing that the transformer achieves the lowest error among all compared models.

are randomly sampled to ensure broad waveform diversity; the complete parameter ranges and sampling distributions are detailed in SI−S1. The final synthetic corpus comprised 60,000 paired CV−DPV samples in total, corresponding to 10,000 samples for each of the six mechanisms.

To reduce dependence on a single potential window, the potential window is randomized on a per-sample basis. Using the sampled window, we generate a CV as the model input. The DPV target is generated on a fixed potential grid aligned with the forward scan axis, ensuring a consistent target length and a well-defined correspondence between encoder potentials and decoder query positions.[11,12] The CV/DPV protocol definitions (including the base−pulse difference rule), discretization, and numerical simulation procedures are described in Supporting Information S2.

For model input, the CV is encoded as a fixed-length token sequence constructed from sampled potential and current values; in the default setting, each token is $[E_i, I_i]$. To distinguish the forward and reverse scan branches in full-cycle inputs, each token additionally carries a scan-direction indicator, which is provided to the model as a categorical attribute (implemented via a learnable embedding). In an ablation setting, we augment the CV token with a local slope feature (dI/dE); its finite-difference definition and numerical safeguards are provided in Supporting Information S3. On the output side, the decoder receives the DPV grid potentials as query positions and predicts the corresponding DPV current values at those positions. Finally, to emphasize waveform morphology rather than absolute current scale, we apply per-sample normalization using a single scale factor computed from each paired CV−DPV sample; the normalization scheme and

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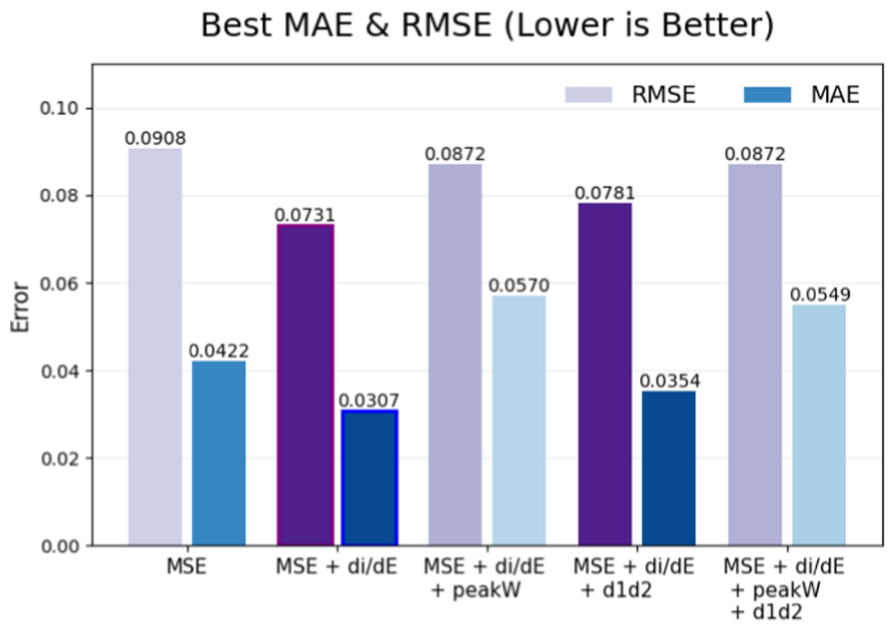


Figure 2. Effect of loss-function design on CV−to−DPV translation accuracy. Best validation MAE and RMSE achieved by models trained with different loss formulations (MSE alone vs MSE augmented with derivative-based penalties and peak-weighting terms), highlighting that incorporating dI/dE and higher-order derivative constraints improves predictive performance compared with MSE-only training.

complete preprocessing protocol are provided in Supporting Information S3. Using these protocol-defined paired data and representations, we train a query-based transformer and evaluate performance as described below.

3.3. Model Training and Evaluation

Using the input/output representations defined above, we employ a query-based transformer for CV → DPV translation (Scheme 2).[15−17] The encoder maps the tokenized CV sequence $\{u_i\}_{i=1}^{S}$ into a contextual latent representation (memory) $\mathbf{M} \in \mathbb{R}^{S\times d}$ via stacked self-attention and feed-forward layers. The decoder takes the DPV potential grid $\{E_j^*\}_{j=1}^{T}$ as query inputs, embeds each grid potential into a query vector q_j, and predicts the DPV current at each grid point by cross-attending to the encoder memory (i.e., $Q = q_j$ and $K, V = M$). This formulation is well suited to waveform translation because it combines global context modeling over the full CV with pointwise retrieval of the most informative CV regions for each DPV output coordinate.

Models are trained to minimize the mean squared error (MSE) between the predicted and reference DPV waveforms on the fixed grid. In controlled ablations, we additionally evaluate two optional loss variants: a peak-weighted reconstruction loss that emphasizes high-magnitude DPV regions and a derivative-consistency term that penalizes mismatches in waveform slope/curvature. Exact definitions and coefficients are provided in the Supporting Information S5.

All neural models are trained using a common optimization protocol with mini-batch gradient-based learning and early stopping based on validation performance. To ensure fair comparisons across models and ablations, we use consistent data set splits and fixed random seeds whenever applicable. Architectural hyperparameters and training settings are documented in Supporting Information S4 and S5.

Performance is reported using waveform-level errors on the DPV grid, specifically MAE and RMSE. We conduct (i) benchmarking against multiple baseline model families, (ii) ablations of input features and loss components, (iii) robustness tests involving protocol perturbations, output-grid variations, and practical domain-shift conditions, and (iv) attention-based analysis that visualizes how decoder cross-attention distributes weight across CV regions when producing DPV predictions. Detailed protocols for these studies are provided in Supporting Information S6.

3.4. Electrochemical Experiments

Electrochemical measurements were performed using a computer-controlled potentiostat in a standard three-electrode configuration at ambient temperature. Potentials are reported versus an Ag/AgCl reference electrode, and currents were converted to current densities using the geometric area of the working electrode.

3.4.1. *p*-Benzoquinone (*p*BQ): Paired CV−DPV Measurement (pH 3.6 Buffer). *p*-Benzoquinone 5.0 mM was dissolved in an aqueous buffer adjusted to pH 3.6. CV[24,25] and DPV were recorded at 200 mV/s under matched solution and electrode conditions to form a paired experimental data set for direct comparison between measured DPV and translated DPV. DPV was acquired on a 5.0 mV potential increment over the same potential window used for the corresponding CV (detailed DPV waveform parameters and acquisition settings are provided in the Supporting Information).

3.4.2. Aniline Electropolymerization: CV Series for Virtual-DPV Analysis. Aniline electropolymerization was carried out in an acidic aqueous electrolyte containing 50 mM aniline monomer and 1 M HNO_3 on a Au disk electrode.[26−28] Ten consecutive CV cycles were recorded from −0.10 to 1.20 V at 50 mV/s to capture nucleation and film-growth evolution. Because an independent DPV scan during growth is experimentally impractical as a stationary readout, DPV was not acquired for this system; instead, each measured CV cycle was translated into a model-derived "virtual DPV" for peak-resolved, cycle-by-cycle feature tracking (Section 4.5).

4. RESULTS AND DISCUSSION

4.1. Benchmarking and Model Selection

To select a backbone for CV → DPV translation, we benchmarked representative model families under identical conditions: the same paired CV−DPV data set and train/validation split, the same preprocessing/normalization and fixed DPV target grid, and the same validation metrics (MAE and RMSE; details in Supporting Information S5.5).

As shown in Figure 1, the Transformer achieves the lowest error on both metrics (MAE = 0.0089, RMSE = 0.0209), outperforming the next-best CNN and RNN (MAE = 0.0120 and 0.0136, RMSE = 0.0258 and 0.0352). Convolutional and classical regression baselines are consistently worse, and slope-

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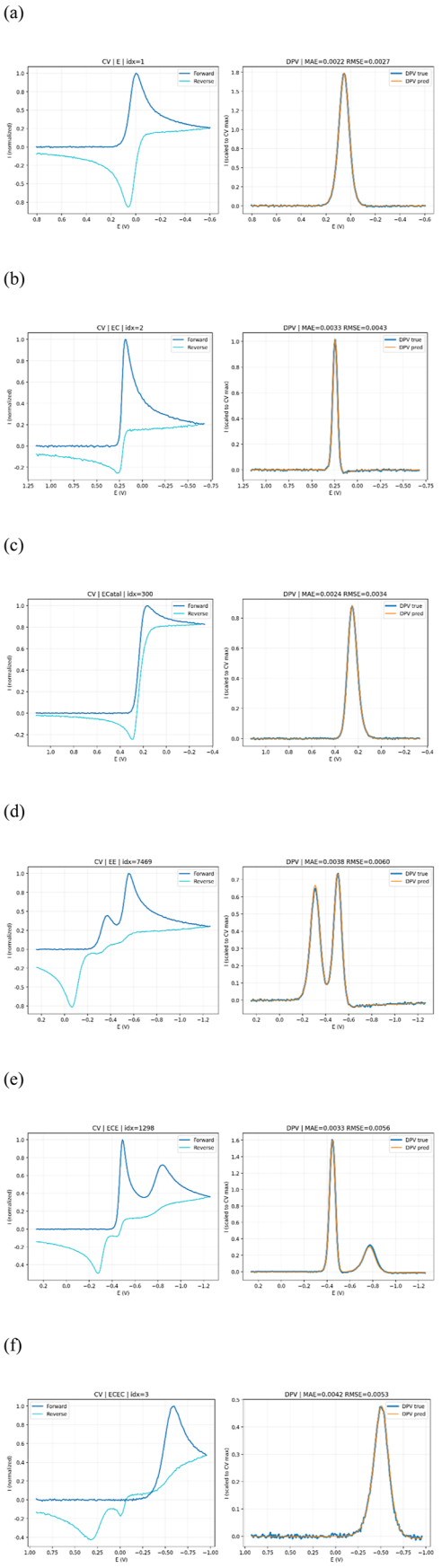


Figure 3. Representative examples of CV−to−DPV translation on the validation set. For each case, the input CV (left; forward and reverse scans) is shown together with the corresponding DPV output (right), comparing the ground-truth DPV with the transformer prediction. The close overlap across single-peak and multipeak responses demonstrates accurate reconstruction of DPV peak positions and amplitudes, with MAE and RMSE reported for each example.

based heuristics that approximate DPV from CV by dI/dE fail by nearly an order of magnitude (MAE >0.04; RMSE >0.1). This negative control matters because it rules out a trivial, purely local mapping; in our setting, accurate translation requires a learned model that can combine local shape cues with broader context along the CV.

We therefore adopt the transformer as the backbone for the remainder of the study. In addition to delivering the best validation accuracy, its query-based decoding with cross-attention provides a direct mechanism to form a soft alignment between each DPV grid point and informative CV regions across the full cycle,[15,17] and it enables architecture-native interpretability via attention inspection (Section 4.4). With this backbone fixed, we next optimize only the minimal input representation and loss choices required to retain accuracy (Section 4.2).

4.2. Ablation-Driven Optimization of the LLM-like Transformer

With the Transformer backbone fixed (Section 4.1), we performed a focused ablation to determine the minimal design choices required for reliable CV → DPV translation. We changed only two factors: (i) the CV token representation (raw current alone vs adding a dI/dE channel) and (ii) the training objective (plain MSE vs additional peak-weighting and/or derivative-consistency terms). All other elements−architecture, data split, normalization, and DPV target grid−were held constant (details in Supporting Information S5.6).

Figure 2 reveals a decisive pattern: performance is primarily limited by representation, not by increasingly complex loss engineering. The baseline model trained with MSE on raw CV tokens shows RMSE = 0.0908 and MAE = 0.0422. Adding a single auxiliary channel, dI/dE, yields the best and most consistent improvement (RMSE = 0.0731, MAE = 0.0307). This result is not merely incremental; it indicates that the model benefits most when the input explicitly exposes the CV features that are most predictive of DPV morphology−slope and inflection cues that mark onset, peak formation, and decay.[29,30] In other words, dI/dE acts as a compact descriptor that explicitly exposes peak-forming transitions and reduces the need for the network to infer them implicitly from absolute current alone.[31]

By contrast, adding constraints to the loss does not improve generalization. Peak-weighting (peak W) substantially degrades accuracy (RMSE = 0.0872, MAE = 0.0570), suggesting that overemphasizing high-current regions disrupts global waveform consistency, including baseline and peak width that ultimately define DPV interpretability. Derivative penalties (d1d2) remain inferior to the simpler dI/dE—only setting (RMSE = 0.0781, MAE = 0.0354), and combining peakW with d1d2 does not recover performance (RMSE = 0.0872, MAE = 0.0549).

We therefore fix dI/dE—augmented input with the standard MSE loss as the final configuration for all subsequent results (Sections 4.6). (Exact loss definitions and coefficients are reported in Supporting Information S5.6).

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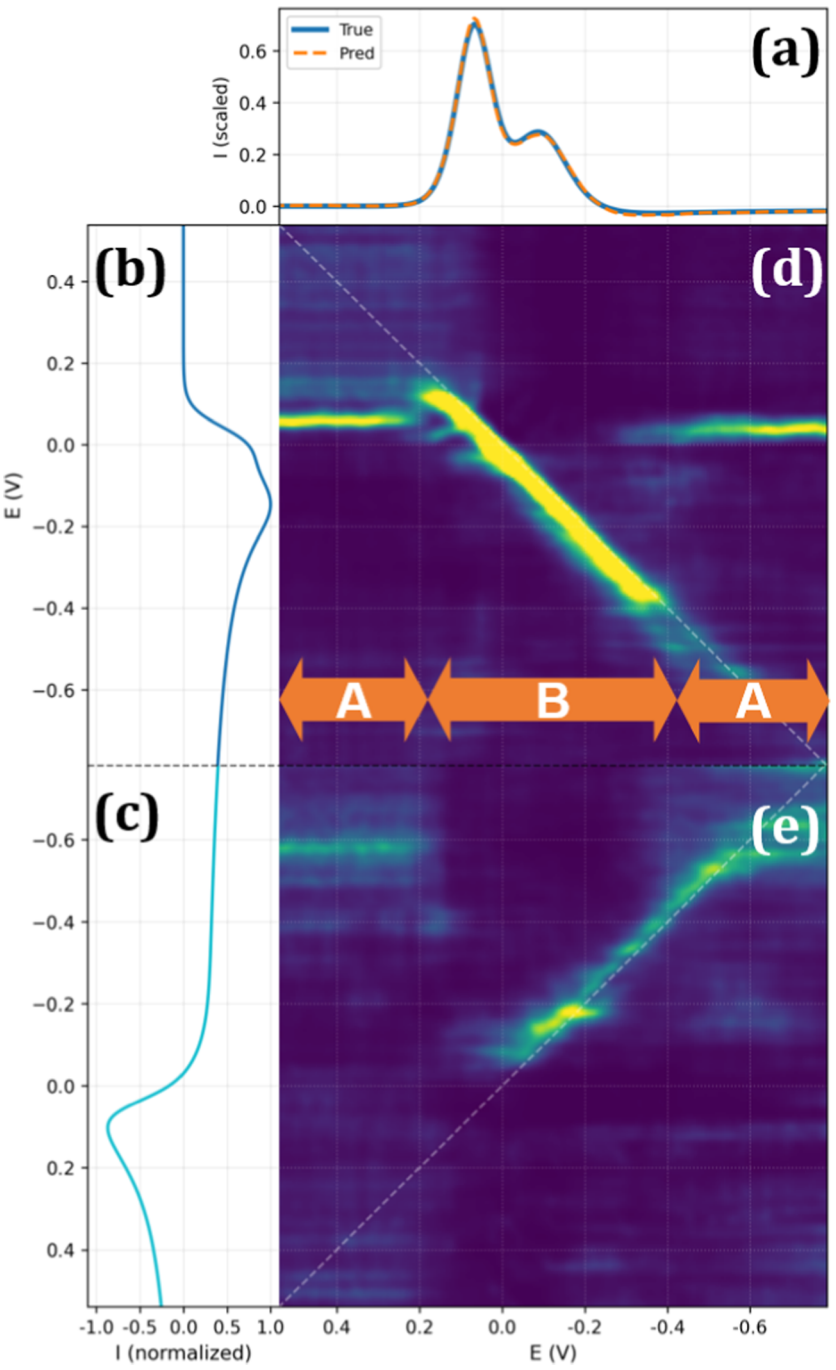


Figure 4. Attention-based interpretation of CV−to−DPV translation for a representative example. (a) Ground-truth and predicted DPV traces. (b,c) Corresponding input CV in the forward and reverse scans [normalized current vs potential, E(V)]. (d,e) Cross-attention map between DPV query potentials (horizontal axis, E) and CV token potentials (vertical axis, E), showing where the model attends on the CV to generate each DPV point. The highlighted regions (A ↔ B ↔ A) indicate distinct attention regimes across the potential range, consistent with the peak and baseline features in the voltammograms.

4.3. Qualitative Waveform Reconstruction Performance Across Six Mechanisms

After fixing the final configuration (Transformer with dI/dE-augmented CV tokens trained under MSE), we test whether the model preserves mechanism-specific DPV morphology rather than merely reducing average error. Figure 3 summarizes representative held-out examples for all six mechanisms (E, EC, ECatal, EE, ECE, and ECEC). Each row shows the full-cycle CV input (left) and the predicted DPV overlaid with the reference DPV on the fixed target grid (right), using the same per-sample scaling for paired CV/DPV.[22]

For the single-step mechanism E, the translator reproduces a single dominant DPV peak with near-complete overlap, indicating accurate reconstruction of both peak position and peak width. The EC and ECatal cases remain well captured despite mechanism-dependent changes in CV curvature and peak sharpness. Importantly, agreement is not confined to the peak apex: the predicted traces track the onset and decay regions

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and maintain a stable baseline, which helps avoid spurious peak-like artifacts in DPV-like outputs.

Multistep mechanisms provide a stricter test because DPV can exhibit multiple features whose spacing and relative amplitudes reflect coupled kinetics and transport. In the EE example, the model reproduces the two-peak structure and the interpeak valley without artificial splitting or merging, which is critical because small errors in peak separation can materially alter interpretation. The ECE and ECEC examples further show that the translator preserves both dominant and secondary peak features and their relative prominence. Across the six representative cases, errors remain consistently low, even as the underlying CV inputs become increasingly complex and history-dependent.

Together, Figure 3 demonstrates that the model does not collapse diverse mechanisms into an "average" DPV; it retains the mechanism-dependent peak structures required for standard peak-based analysis.[9] We next examine how the model achieves this mapping using cross-attention, identifying which CV regions drive each DPV segment during decoding (Section 4.4).

4.4. Explainable AI Analysis: where the Model "Looks" to Reconstruct DPV

To explain how the translator generates DPV from a CV, we analyze cross-attention (Figure 4), which is intrinsic to our query-based decoder: each DPV grid point is a query token that retrieves information from the encoded CV tokens. The attention map therefore reports which CV potentials and scan segments the model uses for each DPV output.

Figure 4 shows an EE example. Panel (a) confirms accurate reconstruction of the two-peak DPV. The attention map (panels d,e separates into two functional regimes along the DPV potential axis (A−B−A). In region A (baseline-dominant DPV potentials), attention collapses onto a narrow, nearly fixed CV potential band (a persistent horizontal stripe), indicating that the model uses an internal "anchor" to stabilize the DPV baseline and suppress spurious peak-like artifacts away from peak windows. In region B (peak-forming potentials), attention becomes structured and query-dependent. A thick near-diagonal ridge appears, indicating soft alignment between DPV queries and CV tokens at similar potentials. Importantly, the ridge is systematically offset by ∼50 mV, comparable to the DPV pulse amplitude, which is consistent with DPV being a base−pulse difference measured across a local potential neighborhood rather than a pointwise current at a single potential. In addition, localized hotspots appear in the reverse potential scan, showing that the model selectively consults scan-history information to resolve multifeature structure (here, the EE two-peak signature).

Overall, Figure 4 suggests a two-mode strategy: baseline anchoring in A and protocol-consistent, locally aligned retrieval (with selective history context) in B. This behavior aligns with Figure 2, where adding dI/dE—a direct cue for transitions and inflection—was the most effective minimal modification.

4.5. Experimental Validation on Real Electrochemical Systems

Sections 4.4 showed that the proposed query-based transformer (MSE objective with the CV slope cue, dI/dE) translates simulated full-cycle CV inputs into DPV waveforms while preserving peak morphology.[32−37] Here we assess whether the same trained translator remains useful on real experimental data, where baseline offsets and evolving interfacial states can deviate from idealized simulations.[2,11,18]

We consider two experimental scenarios. First, *p*-benzoquinone (*p*BQ) in buffered aqueous media (pH = 3.6) provides a paired test of an ECEC-type system, since CV and DPV can be measured under matched conditions for direct comparison.[25,38] Second, aniline electropolymerization represents a DPV-impractical regime: the electrode/film state changes continuously during growth, so pulse-based readouts are difficult to obtain and interpret as stationary signatures.[26−28,39]

For each experiment, the measured CV was resampled onto the model input grid, normalized using the same per-trace scaling as in training, and fed to the trained translator. The model outputs a DPV waveform on the fixed target potential grid.

4.5.1. Paired Validation on pBQ in Buffered Aqueous Solution: an ECEC-Type Example. For the paired pBQ data set, we quantify agreement between translated and measured DPV using (i) peak potential error (ΔE_{peak}), (ii) peak height ratio, (iii) fwhm ratio, and (iv) shape similarity after peak normalization (Pearson r and normalized RMSE).[12,18]

In buffered pH 3.6, we measured CV and DPV under matched conditions and compared the measured DPV with the DPV translated from the corresponding CV (Figure 5a). The translated DPV reproduces the measured morphology across the onset, apex, and decay regions. Quantitatively, ΔE_{peak} is 8 mV, the peak height is 0.75× the experimental value, and the fwhm ratio is 0.88 (slightly narrower). Notably, the experimentally measured DPV exhibits a small offset baseline current, whereas the translated DPV is baseline-free by construction. After baseline correction and peak normalization, the two traces nearly overlap, yielding $r = 0.99$ with normalized RMSE = 0.055. These results indicate that a model trained on synthetic pairs generalizes to experimental data and preserves DPV attributes most relevant for interpretation (peak position, width, and overall shape).

4.5.2. Application to Aniline Electropolymerization: Virtual DPV during Nonstationary Film Growth. Differential pulse voltammetry assumes that the interfacial background is sufficiently stable over the pulse sequence so that baseline subtraction yields a stationary, peak-based readout. During aniline electropolymerization, however, the electrode surface and film properties evolve cycle-by-cycle through nucleation and growth, making an independent DPV scan difficult to obtain and even harder to interpret as a consistent analytical signature. We therefore generate a model-derived DPV ("virtual DPV") directly from the experimentally recorded CV cycles in Figure 6, and assess whether the resulting peaks exhibit coherent, mechanism-consistent evolution.

In this growth regime (cycles 2−10), the virtual DPV consistently resolves two midpotential peaks at peak A around 0.32 V and peak B around 0.47 V, with minimal peak-potential drift ($\sigma_E \approx 8.1$ mV for A; 2.0 mV for B). Gaussian fits further show that the peak widths are essentially cycle-invariant and DPV-like, with fwhm on the order of ∼80−90 mV (i.e., comparable to the reversible DPV half-height width scale fwhm ≈90.5/n mV), corresponding to an apparent $n \approx 1$ width-consistency check rather than a definitive electron stoichiometry for the polymer film.[40] Peak currents, by contrast, change strongly and systematically: peak A increases from 0.499 → 2.101 mA/cm^2 and peak B from 0.495 → 2.065 mA/cm^2 (cycles 2 → 10), with quasi-linear growth slopes of 0.199 and 0.190 mA/cm^2 per cycle ($R^2 = 0.987$ and 0.975). The near-equality of these slopes (ratio ≈ 0.96) indicates that the electroactive populations responsible for peaks A and B scale proportionally

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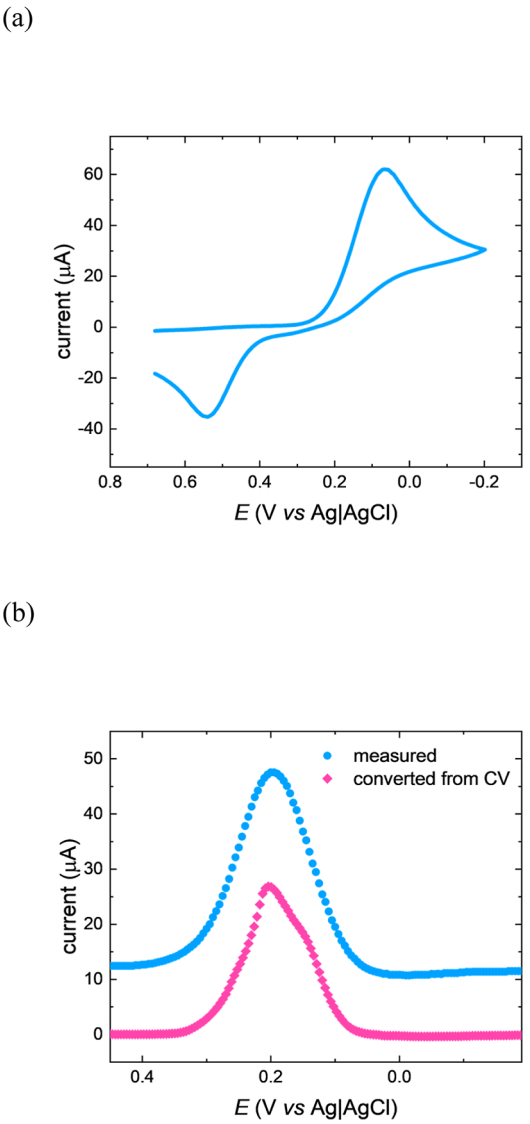


Figure 5. Experimental CV−to−DPV translation for *p*-benzoquinone (*p*-BQ). (a) Measured cyclic voltammogram (CV) of p-BQ used as the model input. (b) Differential pulse voltammograms (DPVs) obtained by direct measurement and by transformer-based translation from the CV, showing agreement in peak position and shape.

during film accumulation, consistent with a conserved film-associated redox/doping manifold that is resolved into two components (e.g., due to film heterogeneity and/or coupled protonation−doping effects) rather than two unrelated, independently evolving species. A higher-potential film peak (peak C around 0.71 V) also grows more gradually (0.280 → 0.771 mA/cm^2, cycles 2−10), consistent with the progressive development of higher-oxidation redox activity as growth proceeds.

Finally, while the raw CV already reflects overall growth, its features remain broad and overlapped under nonstationary electropolymerization conditions. The practical gain of CV → DPV translation is therefore not new chemistry, but a DPV-style, peak-resolved representation in which E_{peak}, fwhm, and cycle-to-cycle peak-current changes can be extracted using consistent criteria across cycles. This makes the translated DPV a practical computational readout for tracking film growth and redox evolution when direct DPV acquisition during polymerization is experimentally impractical.

4.6. Robustness to Protocol and Sampling Variations

To examine whether the CV → DPV translator remains useful beyond the exact base training configuration, we performed additional robustness tests involving protocol perturbations, output-grid variations, and practical domain-shift conditions. Because the base synthetic data set was constructed under a fixed scan rate, a fixed pulse-time fraction, a fixed pulse amplitude, and a decoder query grid aligned with the input CV, these analyses are intended to assess robustness around the base setting rather than to establish full protocol invariance. Detailed protocols, representative examples, and quantitative summaries are provided in the Supporting Information.

First, protocol sensitivity is examined by varying pulse amplitude and scan rate. When the model trained at 50 mV is applied without retraining to data sets generated at 25, 75, and 100 mV, prediction is best at the training condition and degraded as the pulse amplitude departed from 50 mV. This behavior is electrochemically reasonable because changing the pulse amplitude alters both the magnitude and the shape of the DPV response. By contrast, when the model is evaluated on data sets generated at scan rates of 100, 200, 400, and 800 mV/s, the errors remains small and representative overlays show close agreement throughout the tested range. Thus, within the tested range, changes in scan rate has only a small effect on prediction accuracy, whereas changes in pulse amplitude has a much stronger effect.

Second, robustness to the definition of the DPV output grid is examined. Although the default setting uses a 401-point full-cycle CV and a 201-point DPV target aligned with the CV's forward branch, the translator remains effective under both coarser and finer discretizations and the present grid composed of 401/201 provides a practical compromise between numerical accuracy and computational cost. Additional tests with modestly jittered DPV query positions shows that the translated waveforms still preserves the overall peak shape, indicating that performance is not explained solely by exact point-to-point coordinate matching. Likewise, when the output grid is prescribed independently from the original CV discretization at multiple target lengths, the translated DPVs retain the main waveform structure, although denser output grids are numerically more favorable. Therefore, these results support partial flexibility with respect to output-grid definition, rather than strict dependence on a single matched discretization.

Third, practical domain-shift conditions are considered which can be encountered by nonideal electrochemical measurements. Under simulated *iR* drop, the translated and target DPVs remains in good agreement throughout the tested range, with noticeable deviation appearing only at the highest perturbation. Similarly, increasing the capacitive background progressively distorts the CV baseline, but the translated DPV remains reliable over a broad range and deteriorates mainly under the most extreme capacitance values. Because the model is trained on per-sample normalized CV−DPV pairs, its primary task is waveform translation rather than direct prediction of the absolute current scale. Nevertheless, after restoring the original sample-specific current scale, the translated DPVs remains in close agreement with the targets under the tested conditions, indicating that quantitative consistency is preserved reasonably well after rescaling even though the model does not predict the scale on its own.

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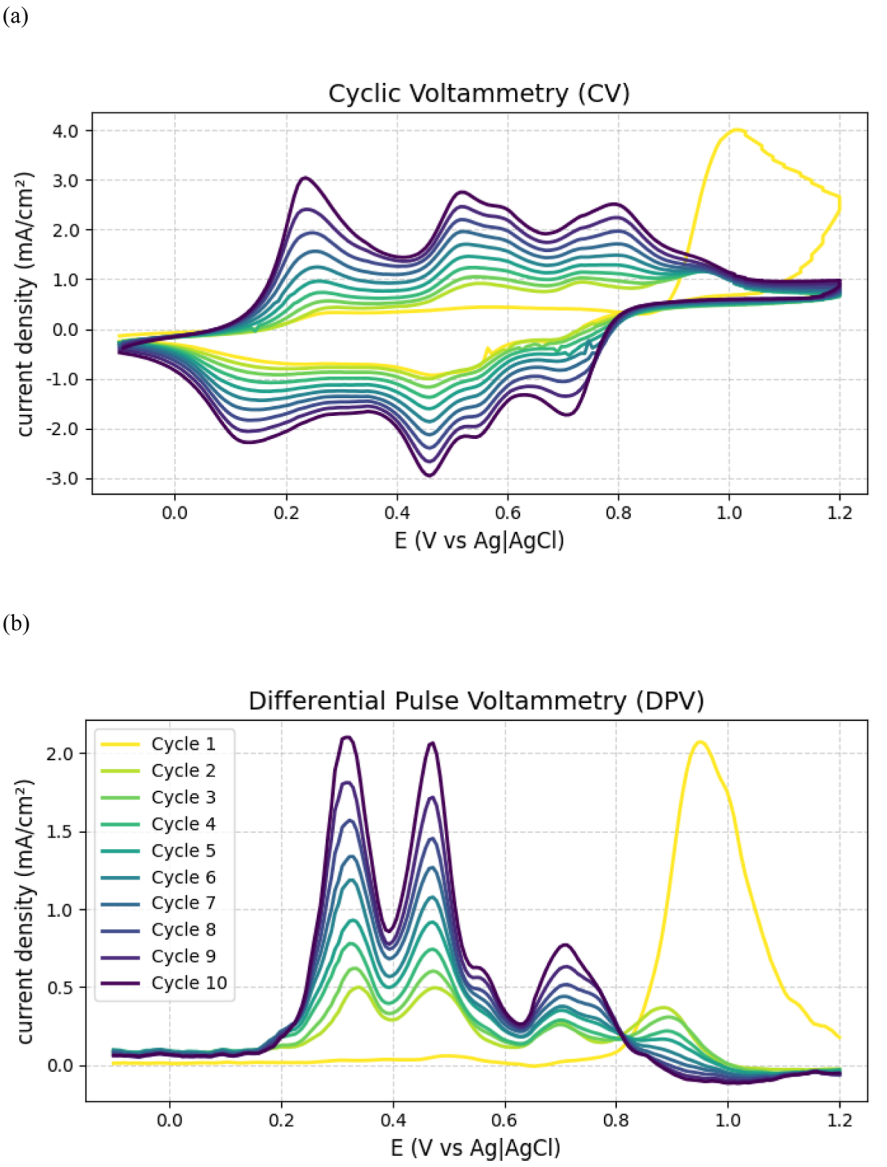


Figure 6. Virtual DPV tracking during electrochemical polymerization. (a) Experimentally recorded CVs over 10 consecutive polymerization cycles (current density vs potential, Ag/AgCl). (b) Corresponding virtual DPVs generated by transformer-based translation of each CV, enabling cycle-by-cycle monitoring of the evolving redox features without performing DPV measurements at every cycle.

Finally, to probe extrapolation beyond the six mechanisms used for training, the pretrained translator is applied to CVs generated from a previously unseen CE mechanism. The predicted DPVs remains in reasonably good agreement with the CE targets and reproduced the main peak positions and overall waveform shapes without severe failure. We do not interpret this result as demonstrating universal mechanism-independent generalization; however, it does indicate that the present framework is capable of meaningful extrapolation beyond the original mechanism library.

Taking all the robustness tests into consideration, it can be concluded that the present translator is best viewed as a fixed-protocol proof-of-concept model with nontrivial practical robustness. The framework tolerates several perturbations around the base setting, especially in scan rate, output-grid definition, moderate *iR* drop, and moderate capacitive-background variation, but it remains sensitive to key protocol variables such as pulse amplitude and does not yet provide universally protocol-independent or scale-aware prediction.

4.7. Limitations and Outlook

Because the translator was trained on paired CV−DPV data sets generated under a fixed DPV protocol (E_{pulse} = 50 mV, pulse-width ratio = 4:6), the converted DPV should be interpreted as the waveform expected under that same measurement condition.[18,41] As discussed in Section 4.6, the model shows useful robustness to several perturbations around the base setting, but its quantitative accuracy still depends on key protocol and data conditions, particularly pulse amplitude and broader domain shifts.[42,43] Accordingly, the present framework should be regarded as a fixed-protocol proof-of-concept rather than a universally protocol-invariant translator. In addition, the availability of paired experimental CV−DPV data sets remains limited, and differences in instrumentation, preprocessing, acquisition settings, data formats, and reporting practice across laboratories may further shift the waveform distribution seen by the model. Broader deployment will therefore likely require protocol- and laboratory-aware adaptation, potentially through

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protocol-conditioned models, small paired calibration sets, or laboratory-specific fine-tuning.[14,41,44]

Future work will therefore focus on (i) constructing paired data sets spanning multiple DPV parameter sets and explicitly reporting the acquisition metadata required for cross-study reuse, and (ii) developing a protocol-aware translator that conditions generation on DPV settings (or is efficiently retrained per setting) using conditional sequence modeling. Extending the same translation framework to other pulse techniques (e.g., SWV/NPV) under clearly specified acquisition conditions is also a natural next step.[9−11,21,45]

5. CONCLUSIONS

This work aimed to cast voltammogram translation as a protocol-to-protocol sequence regression problem and demonstrates that a query-based transformer can translate full-cycle CV inputs into DPV waveforms while preserving DPV-relevant morphology across diverse simulated mechanisms and in representative experimental tests (Figures 1, 3, and 5).

More broadly, the significance of this study lies in establishing voltammogram translation as a practical framework for electrochemical interoperability. Analogous to how transformer-based models translate between languages while preserving meaning, different voltammetric protocols can be treated as distinct "dialects" that encode the same underlying electrochemical information in different waveform forms; learning a protocol-to-protocol mapping therefore makes it possible to compare and harmonize data sets acquired under different excitation methods, rather than restricting analysis to protocol-specific pipelines (Figures 1 and 4). In this view, "translation" is operationally an alignment-and-aggregation rule: the model uses attention to retrieve the CV regions most informative for reconstructing each DPV point on the target grid—potentially leveraging scan-history cues when they matter—thereby enabling a principled conversion between measurement modalities (Figure 4). Practically, such a translator can support cross-protocol data set harmonization and provide virtual pulse readouts when direct pulse acquisition is impractical, while remaining complementary to mechanistic interpretation rather than substituting for it (Figure 5 and 6).

Although the present study focused on CV-to-DPV translation, the same framework may be extendable to related pulse techniques such as square-wave voltammetry (SWV). DPV was intentionally selected here as a stringent proof-of-concept target among pulse-based voltammetric methods. The successful reconstruction of DPV therefore suggests that translation to other pulse waveforms may also be feasible. Yet, this possibility still needs to be verified, because each technique has its own acquisition and readout structure. An important next step will be made to expand the present framework toward multiprotocol translation, including SWV, so that multiple pulse-voltammetric responses may be inferred from a single CV for broader electrochemical interpretation and mechanistic analysis.

■ ASSOCIATED CONTENT

Data Availability Statement

The code used for synthetic data generation and machine-learning training/inference will be made available at the GitHub repository CVtoDPV: https://github.com/Byoung-Yong/CVtoDPV. Because the training corpus is fully synthetic and generated from the simulation rules and parameter ranges described in the Supporting Information, the data set can be regenerated from the released code.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsmeasuresciau.6c00048.

The Supporting Information provides methodological details, technical information, and supplementary analyses. It includes the synthetic mechanism set and parameter sampling strategy (SI−S1); voltammetric protocols and numerical simulation procedures for paired CV−DPV data generation (SI−S2); digital preprocessing steps such as normalization, scan-direction encoding, and optional dI/dE construction (SI−S3); the transformer architecture and hyperparameter settings (SI−S4); training details, loss definitions, and baseline model specifications (SI−S5); experimental acquisition parameters for the *p*-benzoquinone and aniline electropolymerization studies (SI−S6); and additional robustness and generalization analyses, including protocol perturbations, output-grid variations, nonideal electrochemical conditions, scale restoration, and extrapolation to an unseen mechanism (SI−S7) (PDF)

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Notes

The author declares no competing financial interest.

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